Improving Explainable Chart Understanding in Vision-Language Models

Suhyeon Yu (sy118) Yeonju Kim (yk86) Yu-Tzu Chang (yc222)

Abstract

Chart question answering remains challenging for Visual-Language Models (VLMs) because it requires both precise visual extraction and numerical reasoning. In this project, we investigate two resource-efficient techniques to improve performance without retraining the entire network: inference-time Chain-of-Thought (CoT) prompting and Parameter-Efficient Fine-Tuning via Low-Rank Adaptation (LoRA). We evaluate performance using exact match and qualitative error analysis on ChartQA dataset, with a focus on both answer accuracy and interpretability.

1. Introduction

Chart question answering remains challenging for Visual-Language Models (VLMs) because it requires accurate reading of axes, legends, labels, and values together with numerical reasoning. Since charts encode structured quantitative information, small perception errors can directly lead to wrong answers.

We study two resource-efficient ways to improve chart understanding without retraining the full model: inference-time Chain-of-Thought (CoT) prompting and Low-Rank Adaptation (LoRA) fine-tuning. Using ChartQA, we compare a zero-shot baseline, a zero-shot CoT variant, and a LoRA-adapted model, and evaluate them with exact match and qualitative error analysis to understand when reasoning or lightweight adaptation helps. Figure 1 summarizes the workflow.

2. Related Work

Chart question answering is harder than standard visual question answering because models must recover chart structure and perform numerical reasoning. Benchmarks such as DVQA, PlotQA, and ChartQA highlight the need for chart-specific evaluation [2, 5, 4].

Prior work has explored stronger multimodal models, Chain-of-Thought prompting, and parameter-efficient adaptation. We focus on CoT and LoRA as lightweight ways to improve chart understanding [3, 7, 1].

3. Methodology

We design an experimental pipeline for chart question answering with explainable reasoning.

Task setup. Given a chart image and a natural language question, the model predicts a short textual answer, such as a number, label, or phrase.

Dataset. We use the ChartQA dataset with standard train, validation, and test splits. The training split is used for adaptation, the validation split is used for development, and the test split is used for final evaluation.

Baseline. We first evaluate the pretrained Qwen2-VL-2B model [6] in direct-answer mode. The model receives the chart image and question and is asked to return only the final answer.

CoT reasoning. We then add a reasoning-oriented prompt so that the model first identifies relevant chart structure and values, then performs the required comparison or computation before returning the final answer. Only the extracted final answer is used for evaluation.

Fine-tuning. We apply lightweight LoRA adaptation to the baseline model to assess whether fine-tuning improves chart-specific visual grounding and numerical reasoning.

Comparison settings. We compare:

- Baseline (direct answering) vs. CoT prompting
- Baseline (Zero-shot) vs. LoRA fine-tuned model
- Accuracy gains vs. interpretability gains

4. Experimental Settings

We use a unified setup to ensure fair comparison across all configurations.

Model configuration. We use Qwen2-VL-2B-Instruct for all experiments so that differences reflect direct answering, CoT prompting, and lightweight adaptation rather than backbone changes.

Data usage. We use the training split for lightweight adaptation, the validation split for prompt development and intermediate comparison, and the test split for final evaluation.

Inference settings. We keep preprocessing and decoding consistent whenever possible. The baseline uses shorter

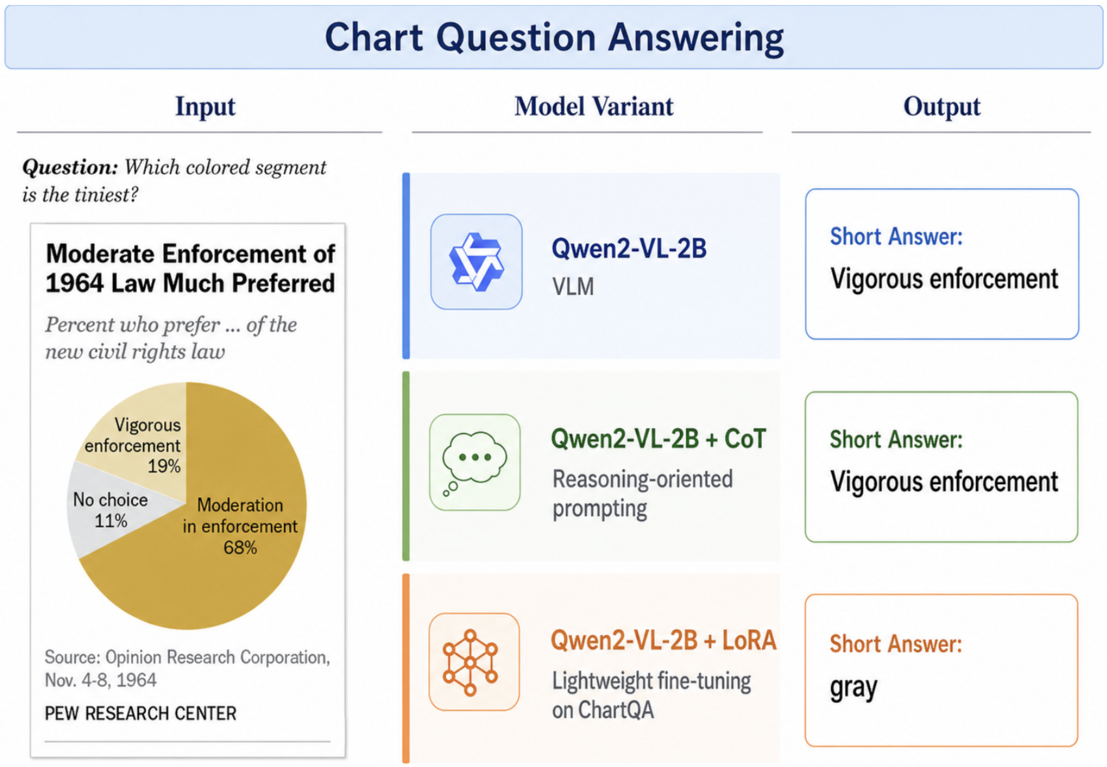


Figure 1. Overview of the three settings compared in this project: zero-shot baseline, zero-shot CoT prompting, and LoRA fine-tuning. All settings use the same base VLM and are evaluated on ChartQA with exact match and qualitative analysis.

outputs, while CoT allows longer generation for intermediate reasoning. All inference is run with batch size 1.

Reasoning setting. In the CoT configuration, the model generates intermediate reasoning before the final answer. We also test a routed variant that applies CoT only to harder question types, such as counting, aggregation, comparison, and arithmetic reasoning.

Implementation details. All experiments are implemented in Python with PyTorch and Hugging Face Transformers. The Qwen2-VL processor prepares image-text inputs, and generated outputs are post-processed to extract the final short answer.

LoRA setting. We use LoRA adapters with rank r=8, α=16, dropout 0.05, and no bias term on the transformer projection layers q_{proj}, k_{proj}, v_{proj}, and o_{proj}, while freezing the vision tower. This yields about 2.18M trainable parameters, or roughly 0.985% of the full model. We train for 1 epoch with batch size 1, gradient accumulation 8, learning rate 2×10^{-4}, maximum sequence length 1024, and gradient checkpointing enabled.

Evaluation protocol. We report Exact Match (EM) as the main metric and supplement it with qualitative success and failure analysis.

5. Results

Overall accuracy. On the ChartQA test set, the baseline model achieves **47.76%** accuracy, the CoT model achieves **45.68%**, and the LoRA-adapted model achieves **52.08%**. Among the three settings, LoRA provides the strongest overall performance, improving over the baseline by **3.92** EM points.

Head-to-head comparison for CoT. On the test set, both models answer **544** examples correctly. The baseline model correctly answers **53** additional examples that the CoT model misses, while the CoT model correctly answers **27** examples that the baseline gets wrong. Both models fail on **626** examples.

These results show that the baseline remains more reliable overall, although the CoT model still provides some complementary improvements on a smaller subset of questions.

Head-to-head comparison for LoRA. On the test set, the baseline and LoRA models both answer **560** examples correctly. The baseline model correctly answers **37** additional examples that LoRA misses, while the LoRA model correctly answers **86** examples that the baseline gets wrong. Both models fail on **567** examples.

These results show that LoRA not only improves the overall EM score, but also solves substantially more questions beyond the baseline, giving a net gain of **49** correct answers.

6. Analysis and Discussion

6.1. Performance Analysis

Baseline model. The baseline model is most effective when stability matters. Its answers are usually short, direct, and aligned with the expected format, which helps on direct lookup, precise numeric reading, simple counting, and straightforward comparisons. Because it does not generate long reasoning chains, it is less likely to drift or introduce formatting errors.

Its main weakness is multi-step reasoning. When a question requires combining several chart values, computing a sum or median, or reasoning across multiple points, the baseline often fails because the answer cannot be obtained from a single lookup.

CoT model. The CoT model helps on some multi-step questions, especially those involving summation, threshold comparison, and counting across multiple points. It also improves interpretability because the intermediate reasoning makes predictions easier to inspect.

However, CoT is less stable. Intermediate steps often introduce extraction errors, incorrect arithmetic, or final answers that do not match the required format. This suggests that the current CoT design is promising, but its prompt and answer control are not yet tight enough.

LoRA model. The main advantage of the LoRA model is better chart-specific grounding. Compared with the baseline, it improves both numeric and text-or-label predictions, as illustrated in Figure 3. Qualitatively, LoRA more often selects the correct year, category, series, or segment value.

Its main weakness is answer-format stability. Under strict exact-match evaluation, LoRA can lose points when it outputs extra characters, multiple answers, or strings that are semantically correct but not normalized to the expected format. Better answer normalization or stricter answer-only constraints could therefore improve measured EM.

Project resources. The code, outputs, and supplementary results for this project are available at `https://drive.google.com/drive/folders/13NWYNz45q03iP99oNx70asDtg3rXdhDx?usp=drive_link`.

6.2. Case Examples

Baseline vs. CoT

We present four representative examples from the test set in Figure 2. The first two favor the baseline, while the last two show where CoT helps on multi-step reasoning.

Baseline vs. LoRA

We present four representative examples from the test set in Figure 3. The first two show stronger baseline stability under strict exact-match evaluation, while the last two show LoRA's better chart-specific grounding.

6.3. Discussion and Potential Improvement

Our current experiments isolate the effects of inference-time CoT prompting and lightweight LoRA fine-tuning, but two extensions appear especially promising. First, a two-stage CoT pipeline could separate rationale generation from final answer prediction and reduce formatting conflicts. Second, because ChartQA does not provide ground-truth reasoning steps, our LoRA adapters are trained only on direct answers. If high-quality reasoning annotations become available, future work could fine-tune reasoning itself through CoT distillation, combining LoRA's efficiency with CoT's interpretability.

7. Conclusion

We compare inference-time CoT prompting and parameter-efficient LoRA fine-tuning for chart question answering in Vision-Language Models. On ChartQA, LoRA performs best with **52.08%** exact-match accuracy, outperforming the baseline (**47.76%**) and CoT (**45.68%**). This suggests that lightweight adaptation improves chart-specific grounding more reliably than prompting alone.

CoT remains useful for interpretability and some reasoning-heavy questions, but its gains are limited by unstable intermediate reasoning and answer-format errors. Overall, LoRA provides the strongest practical improvement, while future work should combine adaptation and reasoning through better answer normalization, two-stage pipelines, and supervised reasoning traces when available.

AI Acknowledgment. During this project, we used Cursor AI as a coding assistant to support code checking, debugging, and refactoring. Its assistance was primarily used to review implementation logic, suggest structural improvements, and help identify potential issues in the code. All final decisions, experiments, analyses, and written content were completed and verified by the author.

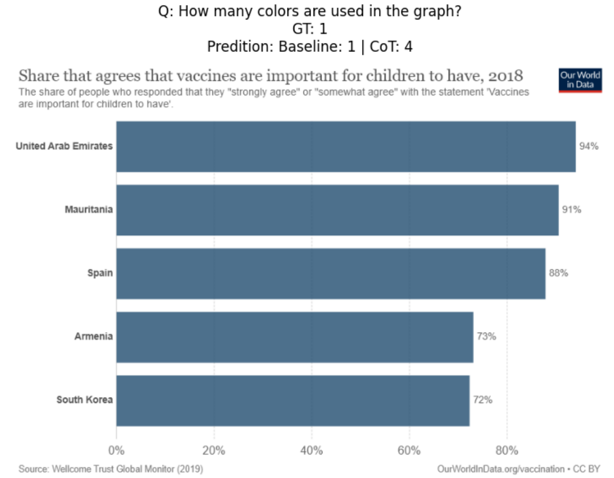
References

[1] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen. Lora: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2022.

[2] K. Kafle, B. Price, S. Cohen, and C. Kanan. Dvqa: Understanding data visualizations via question answering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.

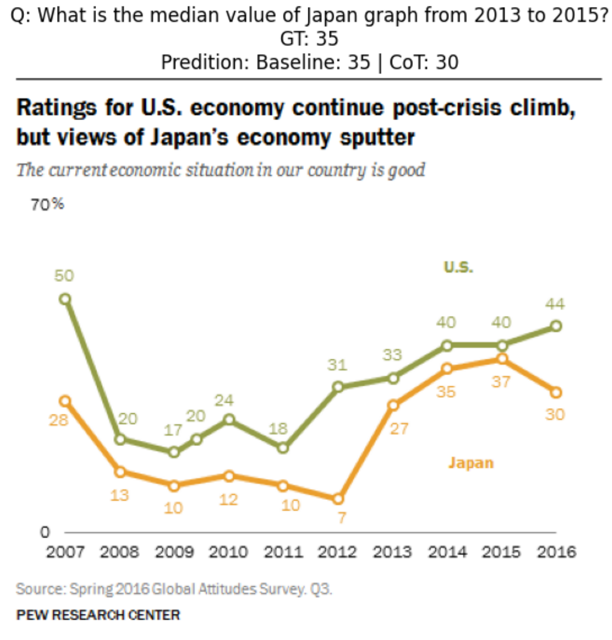
[3] H. Liu, C. Li, Q. Wu, and Y. J. Lee. Visual instruction tuning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[4] A. Masry, D. X. Long, J. Q. Tan, S. Joty, and E. Hoque. Chartqa: A benchmark for question answering about charts



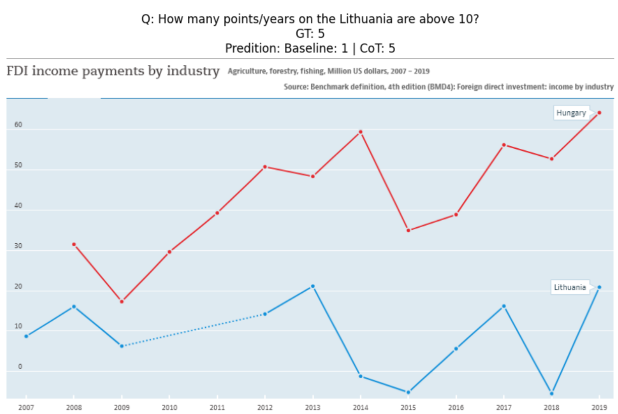
(a) Baseline success. Baseline: *1*; CoT: *4*.

The baseline is more stable on simple lookup questions.



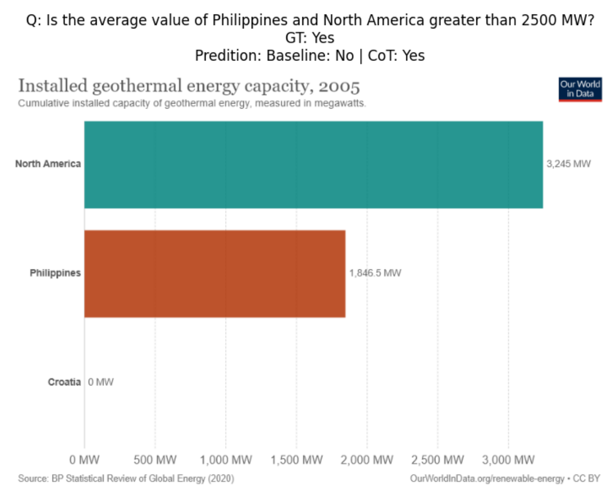
(b) Baseline success. Baseline: *35*; CoT: *30*.

The baseline is more reliable for exact numeric reading.



(c) CoT success. CoT: baseline: *1*; *5*.

CoT helps when multiple points must be counted.



(d) CoT success. baseline: *No*; CoT: *Yes*.

CoT helps with multi-step threshold reasoning.

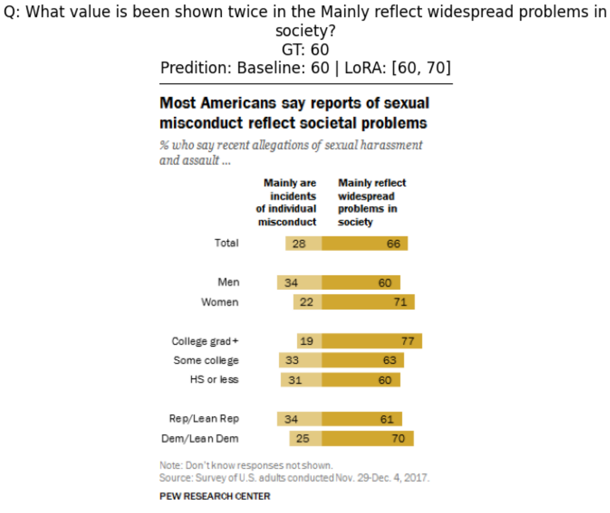
Figure 2. Representative case examples comparing baseline and CoT performance. The top row shows cases where the baseline is more stable and accurate, while the bottom row shows cases where CoT is more effective for multi-step reasoning.

with visual and logical reasoning. In *Findings of the Association for Computational Linguistics: ACL*, 2022.

[5] N. Methani, P. Ganguly, M. M. Khapra, and P. Kumar. Plotqa: Reasoning over scientific plots. In *Proceedings of the IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2020.

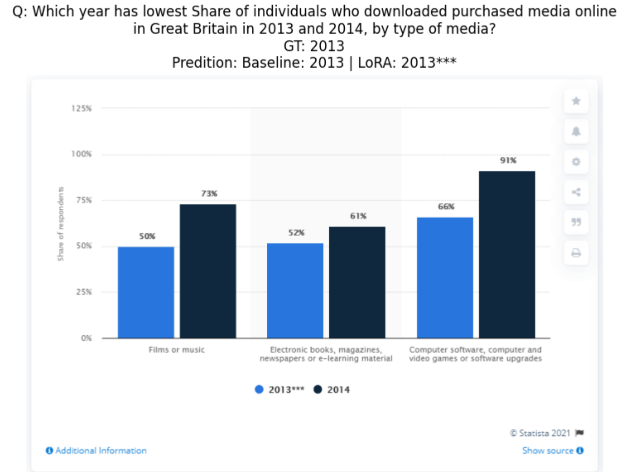
[6] P. Wang, S. Bai, S. Tan, S. Wang, Z. Fan, J. Bai, K. Chen, X. Liu, J. Wang, W. Ge, Y. Fan, K. Dang, M. Du, X. Ren, R. Men, D. Liu, C. Zhou, J. Zhou, and J. Lin. Qwen2-vl: Enhancing vision-language model's perception of the world at any resolution. *arXiv preprint arXiv:2409.12191*, 2024.

[7] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. H. Chi, Q. V. Le, and D. Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.



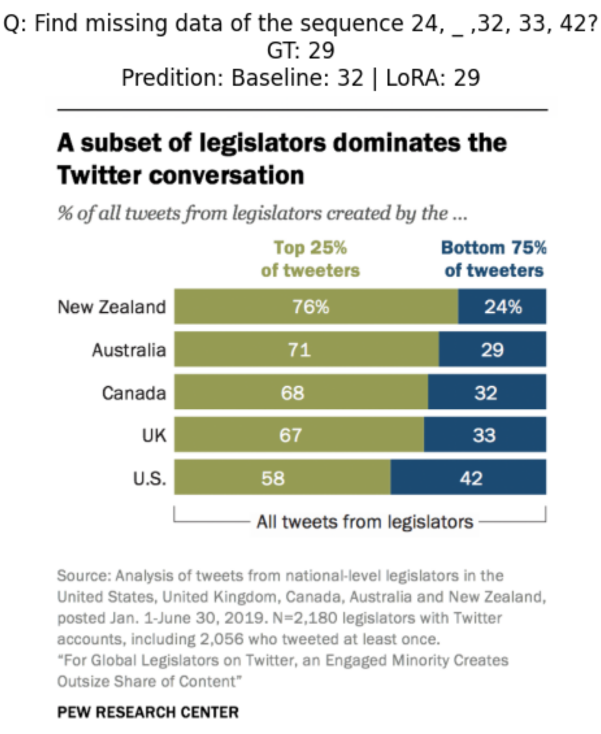
(a) Baseline success. Baseline: *60*; LoRA: *[60, 70]*.

LoRA identifies the relevant values but returns multiple answers, so the baseline is better aligned with strict exact-match evaluation.



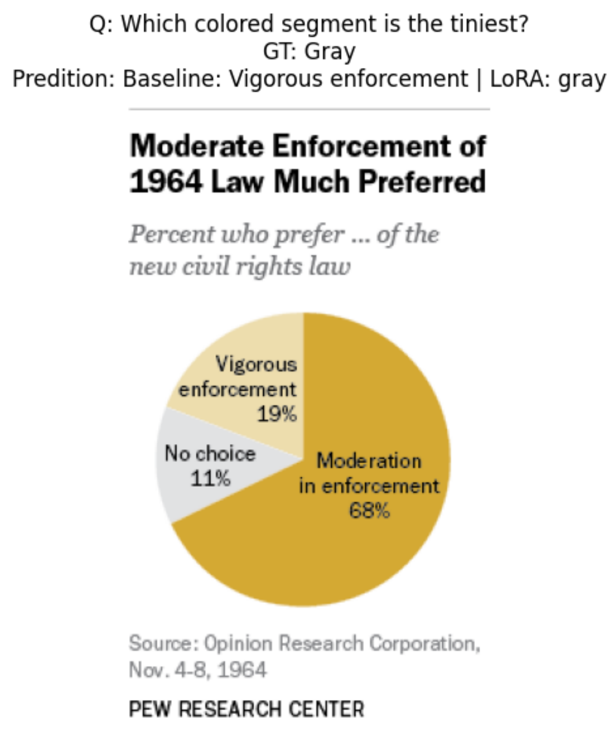
(b) Baseline success. Baseline: *2013*; LoRA: *2013***.

The baseline is more reliable when the task requires a clean, exactly formatted final answer.



(c) LoRA success. Baseline: *32*; LoRA: *29*.

LoRA better grounds the chart structure and selects the correct value instead of a nearby distractor.



(d) LoRA success. Baseline: *Vigorous enforcement*; LoRA: *gray*.

LoRA more accurately maps the visual cue in the chart to the correct target label.

Figure 3. Representative case examples comparing baseline and LoRA performance. The top row shows baseline wins caused by stronger answer-format stability, while the bottom row shows LoRA wins driven by better chart-specific grounding and label/value selection.